Scalability & Future Plan

Date	03 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

The current version of OptiCrop helps users identify suitable crops by analyzing soil nutrients and environmental conditions. As more users and agricultural data become available, the system can be expanded to support larger datasets, real-time services, and additional farming features to improve decision-making and usability.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Uses a static crop dataset for predictions instead of real-time agricultural data.	Prediction accuracydepends on the quality and coverage of the dataset.	**High**
2	Supports crop recommendation and suitability analysis only.	Users cannot receive fertilizer or disease management recommendations.	**Medium**
3	Flask backend is suitable for small to medium traffic but not optimized for large-scale concurrent users.	Response time may increase if many users access the system simultaneously.	**Medium**

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users through the current deployment.	Deploy backend on a scalable cloud platform with load balancing to support more concurrent users.
2	Data Storage	Uses local CSV datasets and trained ML models.	Store agricultural data and prediction history in a cloud database for easier management and scalability.

3	Performance	Prediction is processed using a locally trained machine learning model.	Optimize model loading, API response time, and caching to improve performance under higher workloads.
4	Security	Public web application without user authentication.	Add secure user authentication, HTTPS protection, and role-based access for future versions.

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 1	Integrate real-time weather and soil data APIs.	1 Month	Improve predictionaccuracy using live environmentaldata.
Phase 2	Add fertilizer recommendation based on soil nutrient analysis.	2–4 Months	Help farmers improve soil health and crop productivity.
Phase 3	Implement crop disease detection using image analysis.	4–7 Months	Enable early disease identification and reduce crop losses.
Phase 4	Develop a multilingual mobile application with offline support.	Future Version	Increase accessibility for farmers in different regions and improve usability in low-connectivity areas.